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JEE MAIN PHYSICS DPP: POSITION-TIME GRAPHS

Topic: Graphical Interpretation of Position-Time ($x-t$) Graphs | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** In an $x-t$ graph, the slope of the tangent at any given instant represents:
- Instantaneous acceleration
 - Instantaneous velocity
 - Average velocity
 - Rate of change of acceleration
- Q2.** An $x-t$ graph for a particle moving along a straight line is a parabola opening upwards ($x = kt^2$, $k > 0$). This motion corresponds to:
- Uniform motion with zero acceleration
 - Motion with constant negative acceleration
 - Motion with constant positive acceleration
 - Non-uniformly accelerated motion with increasing acceleration
- Q3.** The $x-t$ graphs of two particles A and B are straight lines making angles of 30° and 60° with the time axis respectively. The ratio of their velocities v_A/v_B is:
- $\sqrt{3} : 1$
 - $1 : \sqrt{3}$
 - $3 : 1$
 - $1 : 3$
- Q4.** If an $x-t$ curve is concave downwards ($\frac{d^2x}{dt^2} < 0$) in a given interval, the particle is experiencing:
- Negative acceleration (retardation)
 - Positive acceleration
 - Zero acceleration
 - Infinite acceleration
- Q5.** Which of the following position-time ($x-t$) graphs is physically **impossible**?
- An $x-t$ graph with negative slope
 - An $x-t$ graph having a vertical tangent (infinite slope)
 - An $x-t$ graph with a horizontal tangent
 - An $x-t$ graph passing below the time axis
- Q6.** The $x-t$ graph of a particle is given by $x = 8t - 2t^2$ m. The particle reverses its direction of motion at time t equal to:
- 4 s
 - 1 s
 - 2 s
 - 0.5 s
- Q7.** If the $x-t$ graph of a moving particle is parallel to the time axis, the state of the particle is:
- Moving with constant positive velocity
 - Moving with constant negative velocity
 - Moving with constant acceleration
 - At rest
- Q8.** A particle moves along the x -axis. Its $x-t$ graph consists of two straight line segments: from (0 s, 0 m) to (2 s, 10 m) and from (2 s, 10 m) to (5 s, 0 m). The average velocity in the interval $t = 0$ to 5 s is:
- 0 m/s
 - 2 m/s
 - 4 m/s
 - 3.33 m/s
- Q9.** In the previous question, the average speed of the particle in the interval $t = 0$ to 5 s is:
- 2 m/s
 - 4 m/s
 - 0 m/s
 - 5 m/s
- Q10.** The position of a particle is $x(t) = t^3 - 3t^2 + 2$ m. The inflection point on the $x-t$ curve (where acceleration is zero) occurs at:
- $t = 0$ s
 - $t = 2$ s
 - $t = 1$ s
 - $t = 3$ s
- Q11.** A ball is thrown vertically upwards from the ground, reaches maximum height, and falls back to the thrower's hand. The correct $x-t$ graph (with upward direction taken as positive) is:
- A straight line with positive slope
 - A straight line with negative slope
 - A parabola opening upwards
 - A parabola opening downwards
- Q12.** Two particles 1 and 2 start from the same point. Their $x-t$ equations are $x_1 = 2t$ and $x_2 = 4t - t^2$. The time at which they meet again after $t = 0$ is:
- 2 s
 - 4 s
 - 1 s

(d) 3 s

Q13. If the tangent to an $x-t$ curve makes an obtuse angle with the positive time axis, the instantaneous velocity is:

- (a) Positive
- (b) Negative
- (c) Zero
- (d) Increasing

Q14. In an $x-t$ graph, the area under the curve:

- (a) Represents total displacement
- (b) Represents velocity
- (c) Has no direct physical significance
- (d) Represents work done

Q15. For a particle executing simple harmonic motion $x = A \sin(\omega t)$, the slope of the $x-t$ graph is maximum at:

- (a) $x = +A$
- (b) $x = -A$
- (c) $x = \pm A/\sqrt{2}$
- (d) The mean position ($x = 0$)

Q16. The position-time graph of a body is shown as a complete circle of radius R . This graph is:

- (a) Physically unrealizable because time cannot flow backwards and one time cannot have two positions
- (b) Valid for uniform circular motion
- (c) Valid for simple harmonic motion
- (d) Valid for oscillatory motion

Q17. A car travels on a straight line. Its $x-t$ curve has $\frac{dx}{dt} > 0$ and $\frac{d^2x}{dt^2} > 0$ throughout. The car is:

- (a) Moving in positive direction with decreasing speed
- (b) Moving in positive direction with increasing speed (speeding up)
- (c) Moving in negative direction with increasing speed
- (d) At rest

Q18. A particle moves according to $x(t) = 5 \sin(2\pi t)$ m. In one full period $T = 1$ s, the number of times the slope of the $x-t$ graph becomes zero is:

- (a) 1
- (b) 4
- (c) 2
- (d) 3

Q19. The $x-t$ graph of a particle is given by $x(t) = a - be^{-kt}$, where a, b , and k are positive constants. As $t \rightarrow \infty$, the velocity of the particle approaches:

- (a) bk
- (b) ak
- (c) ∞
- (d) 0

Q20. An elevator descends from rest at the top floor to the ground floor. Its position $y(t)$ measured from the ground floor as a function of time is best described by:

- (a) An initial downward curved parabola, followed by a linear descent, and ending with an upward curved parabola to rest
- (b) A single continuous straight line with negative slope
- (c) A single parabola opening upwards
- (d) A horizontal straight line

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

By definition, the slope of the position-time curve is:

$$\text{Slope} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = v_{\text{inst}}$$

It gives the instantaneous velocity of the particle.

Sol 2. Correct Option: (c)

For $x(t) = kt^2$ ($k > 0$):

$$v(t) = \frac{dx}{dt} = 2kt.$$

$$a(t) = \frac{dv}{dt} = 2k = \text{constant} > 0.$$

Thus, an upward-opening parabola on an x - t graph indicates uniform positive acceleration.

Sol 3. Correct Option: (d)

$$v_A = \tan(30^\circ) = \frac{1}{\sqrt{3}}.$$

$$v_B = \tan(60^\circ) = \sqrt{3}.$$

Ratio:

$$\frac{v_A}{v_B} = \frac{1/\sqrt{3}}{\sqrt{3}} = \frac{1}{3}$$

Sol 4. Correct Option: (a)

Curvature / concavity is given by the second derivative:

$$\text{Concave downwards} \implies \frac{d^2x}{dt^2} < 0.$$

Since $a = \frac{d^2x}{dt^2}$, the acceleration is negative (retarding motion if moving forward).

Sol 5. Correct Option: (b)

A vertical tangent implies $\frac{dx}{dt} \rightarrow \pm\infty$, which means infinite speed—an unphysical situation. Furthermore, any vertical line segment means a particle exists at multiple positions simultaneously.

Sol 6. Correct Option: (c)

A particle reverses direction when instantaneous velocity crosses zero:

$$v(t) = \frac{dx}{dt} = 8 - 4t = 0 \implies 4t = 8 \implies t = 2 \text{ s.}$$

Sol 7. Correct Option: (d)

If the line is parallel to the time axis, $x(t)$ = constant.

Velocity $v = \frac{dx}{dt} = 0$, meaning the particle is stationary at rest.

Sol 8. Correct Option: (a)

Displacement $\Delta x = x(5) - x(0) = 0 - 0 = 0 \text{ m}$.

Average velocity:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{0 \text{ m}}{5 \text{ s}} = 0 \text{ m/s}$$

Sol 9. Correct Option: (b)

• Segment 1 (0 to 2 s): $d_1 = |10 - 0| = 10 \text{ m}$.

• Segment 2 (2 to 5 s): $d_2 = |0 - 10| = 10 \text{ m}$.

Total distance = $10 + 10 = 20 \text{ m}$.

Average speed:

$$\text{Speed}_{\text{avg}} = \frac{\text{Distance}}{\text{Time}} = \frac{20 \text{ m}}{5 \text{ s}} = 4 \text{ m/s}$$

Sol 10. Correct Option: (c)

Inflection point occurs where concavity changes sign ($\frac{d^2x}{dt^2} = 0$):

$$v(t) = \frac{dx}{dt} = 3t^2 - 6t.$$

$$a(t) = \frac{dv}{dt} = 6t - 6 = 0 \implies t = 1 \text{ s.}$$

Sol 11. Correct Option: (d)

With upward taken as $+y$:

$$y(t) = ut - \frac{1}{2}gt^2.$$

This is a quadratic equation with negative leading coefficient ($-\frac{1}{2}g$), which is a parabola opening downwards.

Sol 12. Correct Option: (a)

For the particles to meet, $x_1(t) = x_2(t)$:

$$2t = 4t - t^2 \implies t^2 - 2t = 0 \implies t(t - 2) = 0.$$

Excluding starting instant $t = 0$, they meet at $t = 2 \text{ s}$.

Sol 13. Correct Option: (b)

Velocity $v = \tan \theta$.

For an obtuse angle ($90^\circ < \theta < 180^\circ$), $\tan \theta < 0$, which signifies a negative velocity (moving backwards along the axis).

Sol 14. Correct Option: (c)

$\int x dt$ has units of metre·second (m·s), which does not correspond to any standard physical quantity.

Sol 15. Correct Option: (d)

$x = A \sin(\omega t) \implies \text{Slope } v = \frac{dx}{dt} = A\omega \cos(\omega t)$.
Maximum slope = $A\omega$, which occurs when $\cos(\omega t) = \pm 1 \implies \sin(\omega t) = 0 \implies x = 0$ (mean position).

Sol 16. Correct Option: (a)

A circle in the x - t plane would require a vertical line to intersect the graph at two distinct positions for a single time instant, and the particle would also move backwards in time on the left semicircle.

Sol 17. Correct Option: (b)

- $\frac{dx}{dt} > 0 \implies$ velocity is positive (moving in $+x$ direction).
- $\frac{d^2x}{dt^2} > 0 \implies$ acceleration is positive.

Since velocity and acceleration have the same sign, the particle is speeding up.

Sol 18. Correct Option: (c)

Slope is zero when velocity $v(t) = 10\pi \cos(2\pi t) = 0$.
For $0 \leq t \leq 1$ s:
 $2\pi t = \frac{\pi}{2}, \frac{3\pi}{2} \implies t = 0.25 \text{ s}, 0.75 \text{ s}$ (exactly 2 times).

Sol 19. Correct Option: (d)

$x(t) = a - be^{-kt}$.
 $v(t) = \frac{dx}{dt} = bke^{-kt}$.
As $t \rightarrow \infty$, $e^{-kt} \rightarrow 0 \implies v(\infty) = 0$.

Sol 20. Correct Option: (a)

- Acceleration phase from rest: downward parabola ($y = H - \frac{1}{2}at^2$).
- Constant velocity descent: straight line with negative slope.
- Deceleration phase to rest: upward curving parabola ($a > 0$ opposes downward velocity).

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